

Data Mining

Sophie Dabo

Cours Moodle : M1 MIASHS : Data Mining (S.Dabo)

Pour s'inscrire sous MOODLE : **bu62ci**

Overview of topics

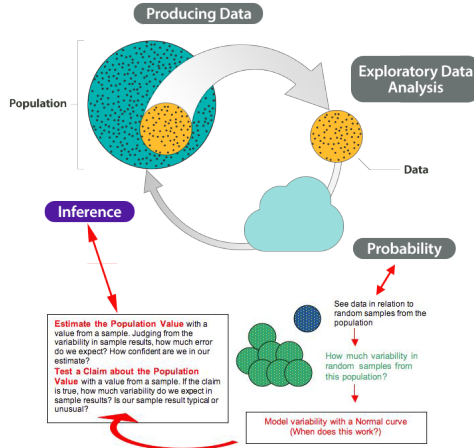
① Part I : Unsupervised-learning

- Introduction
- Clustering
- Applications with [Python](#)

② Part II : Supervised learning

- Linear, Quadratic, Discriminant Analysis,
- Classification : Logistic regression
- Classification : Regression trees, SVM, Bagging, Boosting,...
- Applications with [Python](#)

Why do we need statistics?



Producing data to answer the questions

- **Population** : cannot take measurements of all the individuals (of size M) of the Medical Center.
- **Sample** : take measurements on a subset (of size $N < M$) of individuals from the **Population**.
- **Training Data vs. Test Data** : for modeling, split the sample (often random) into non-overlapping subsets to build and assess the performance of the model
- **Randomisation** is **usual** in biological studies. Some exceptions depend on the population, medical questions/diseases, modelling procedure

According to

- the goal and the type of data of interest :
 - Test
 - Explanatory analysis, Regression
 - Classification
- how the analyses makes use of the data :
 - **Unsupervised** : *select the most informative input variables X for supervised approaches or divide the observations into relatively distinct groups.*
 - **Supervised** : *find (mostly by inference) a mapping function f from a selected set of predictors X to outputs Y*
- the kind of information provided by the statistical methods :
 - **Descriptive** Statistics
 - Statistical **Inference**

Clustering

Aim

Clustering makes it possible to group individuals into a finite number of classes according to their similarities. There are generally two main families of methods :

- **methods based on the optimisation of criteria** such as variance ratios : the method of moving centres (averages) or nearest neighbors.
- **iterative methods** based on groupings of neighbouring individuals based solely on notions of distance (similarity) between individuals : the hierarchical ascending classification method.

Classification of quantitative data

- We observe p quantitative variables $\xi_1, \dots, \xi_p$ on n individuals $X_1, \dots, X_n$.
- The data are therefore presented in the form of an individual-variable table with n rows and p columns.
- X_{ij} is the value of the variable ξ_j observed on the individual named X_i .

Distances, similarity measure for continuous data

- Euclidean : $d(X_i, X_j) = \sqrt{\sum_{k=1}^p (X_{ik} - X_{jk})^2}$
- Minkowski (norm L_m) :

$$d_{ij} = d(X_i, X_j) = \left[\sum_{k=1}^p |X_{ik} - X_{jk}|^m \right]^{1/m}$$

- If we have different scales, we can correct with the variance :

$$d_{ij} = d(X_i, X_j) = \left[\sum_{k=1}^p \frac{|X_{ik} - X_{jk}|^m}{\sigma_k} \right]^{1/m}$$

- Canberra $d(X_i, X_j) = \sum_{k=1}^p \frac{|X_{ik} - X_{jk}|}{X_{ik} + X_{jk}}$
- Czekanowski : $d(X_i, X_j) = 1 - \frac{2 \sum_{k=1}^p \min(X_{ik}, X_{jk})}{\sum_{k=1}^p (X_{ik} + X_{jk})}$

Hierarchical clustering

Hierarchical clustering : Hclust

The iterative Hclust method works as follows :

- **Initialisation** : each individual constitutes a class. The initial partition is then composed of n classes (singletons).
- Calculate the distance matrix (with a chosen relevant distance) 2 by 2.
- Iterate the next two steps until a single class is obtained.
 - i group together the two closest classes in the sense of a distance between groups
 - ii update the distance matrix 2 to 2. Each class is represented by its centre.

Choosing the distance between two classes C_i and C_j

- Minimum jump : $d(C_i, C_j) = \min_{k \in C_i, l \in C_j} d_{kl}$
- Maximum jump : $d(C_i, C_j) = \sup_{k \in C_i, l \in C_j} d_{kl}$
- Average jump : $d(C_i, C_j) = \frac{1}{\text{card}(C_i) + \text{card}(C_j)} \sum_{k \in C_i, l \in C_j} d_{kl}$

where $\text{card}(A)$ is the cardinal (number of elements in the set A).

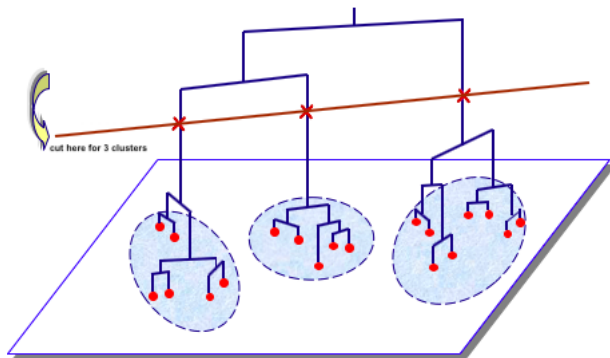
Choosing the distance between two classes C_i and C_j

Let g_i and g_j be the barycentres of the two classes of respective weights w_i , w_j :

- the distance of the barycentres is : $d(C_i, C_j) = d(g_i, g_j)$
- the ward jump is : $d(C_i, C_j) = \frac{w_i w_j}{w_i + w_j} d(g_i, g_j)$ or $(\frac{w_i w_j}{w_i + w_j} d^2(g_i, g_j))$, is linked to a minimisation of the interclass variance growth (between classes).
- **the choice of the number of classes** involves the search for a break in the decrease in the ratio of the intra-class variance to the total variance as a function of the number of classes.
- The **dendrogram** (binary tree) gives the branch height, proportional to the dissimilarity index or distance between two grouped objects.
- A Hclust is often used on the components of a PCA, CA or MCA.

Example of a dendrogram showing a partition into 3 classes

Dendrogram



K-Means Clustering

K-Means Clustering

The k-means method is a dynamic reallocation algorithm based on the maximisation of a global criterion (the ratio of intra-class inertia to inter-class inertia).
maximisation of a global criterion (the ratio of intra-class inertia to inter-class inertia). Unlike Hclust, we choose the number of classes K .

- **Initialisation** : we choose the number of classes K and their centres (K points among n at random).
- Iterate through the following steps until the interclass variance criterion stops varying significantly.
 - i for each individual ($i = 1, \dots, n$) allocate individual i to class k such that $d(X_i, g_k) \leq d(X_i, g_j)$ for all $j \neq k$
 - ii calculate the centres of gravity g_k of the K classes

Note

- The algorithm converges because the criterion cannot increase at each step. The solution obtained is a local optimum.
- The final partition depends on the centres (initial seeds) chosen.
- **The algorithm can be run with different initial seeds**, and the best partition is retained, for example using the class cohesion measure $\sum_{k=1}^K \sum_{i \in C_k} d(X_i, g_k)$.
- Hclust is often used to initialise a moving centre method (to find the number of classes and the centres of the classes).
- If the size of the data is large, perform the Hclust on a random sample of the data.
- The distance is defined according to the type of variables.
- In general, the algorithm uses the Euclidean distance.

Choosing the number of classes

- **Decay of the loss of inter inertia in Hclust** : the search for a jump in the graph of the decay of the inter-class variance in the case of the Ward criteria gives a heuristic indication of the number of classes.
- **Silhouette** : for an individual X_i in a class A (of cardinal ≥ 2) and for a distance $d(.,.)$, X_i is well classified if its silhouette $s(i)$ is large (close to 1),
 $-1 \leq s(i) \leq 1$ let $s(i) = \frac{b(i)-a(i)}{\max(a(i), b(i))}$, with

$$a(i) = \frac{1}{\text{Card}(A) - 1} \sum_{x_j \in A, j \neq i} d(x_i, x_j)$$

$$b(i) = \min_{A \neq C} \frac{1}{\text{Card}(C)} \sum_{x_k \in C} d(x_i, x_k)$$

for all classes C different from A of the same partition.

Choosing the number of classes

- An observation x_i is well classified if $s(i)$ is large (close to 1), $-1 \leq s(i) \leq 1$.
The average $\bar{s}$ of the $s(i)$ is considered to be a global indicator of the quality of a partition.
- **Gap** is another statistics (to maximize) useful for the choice of the number of cluster regarding the homogeneity of the partition

Choosing the number of classes

- **Gap** : let D_r be the sum of all the distances taken between the observations two by two within the same class C_r , $r = 1, \dots, K$. Let W_K be the weighted average (by the size of the class) of these sums of distances.
- The largest deviation in the graph of $\log(W_K)$ as a function of that of a reference distribution obtained by simulation (Monte Carlo) according to a uniform distribution is an overall indicator of the quality of a partition.

Classification of binary data

Let the pairs of observations (X_i, X_j) with $X_i^T = (X_{i1}, \dots, X_{ip})$, with $X_{ik} \in \{0, 1\}$. The measures of similarity are :

$$d_{ij} = \frac{a_1 + \lambda a_4}{a_1 + \lambda a_4 + \mu(a_2 + a_3)}$$

where

$$a_1 = \sum_{k=1}^p 1_{(x_{ik}=x_{jk}=1)}, \quad a_2 = \sum_{k=1}^p 1_{(x_{ik}=0, x_{jk}=1)},$$

$$a_3 = \sum_{k=1}^p 1_{(x_{ik}=1, x_{jk}=0)}, \quad a_4 = \sum_{k=1}^p 1_{(x_{ik}=x_{jk}=0)}$$

where λ, μ are weights which allow similarities or differences to be given greater or less importance. If $\lambda = 0, \mu = 1$ we obtain Jaccard's index. If $\lambda = 1, \mu = 1$, we have the Russel and Rao index.

Similarity between nominal objects

If the objects are nominal, then :

- we refer to a contingency table (we usually use a distance of χ^2)
- or a complete disjunctive table (we use a similarity between objects with binary structures).

To classify categorical data, one could use relevant similarities as above or perform classification on the factorial components of a CA or MCA